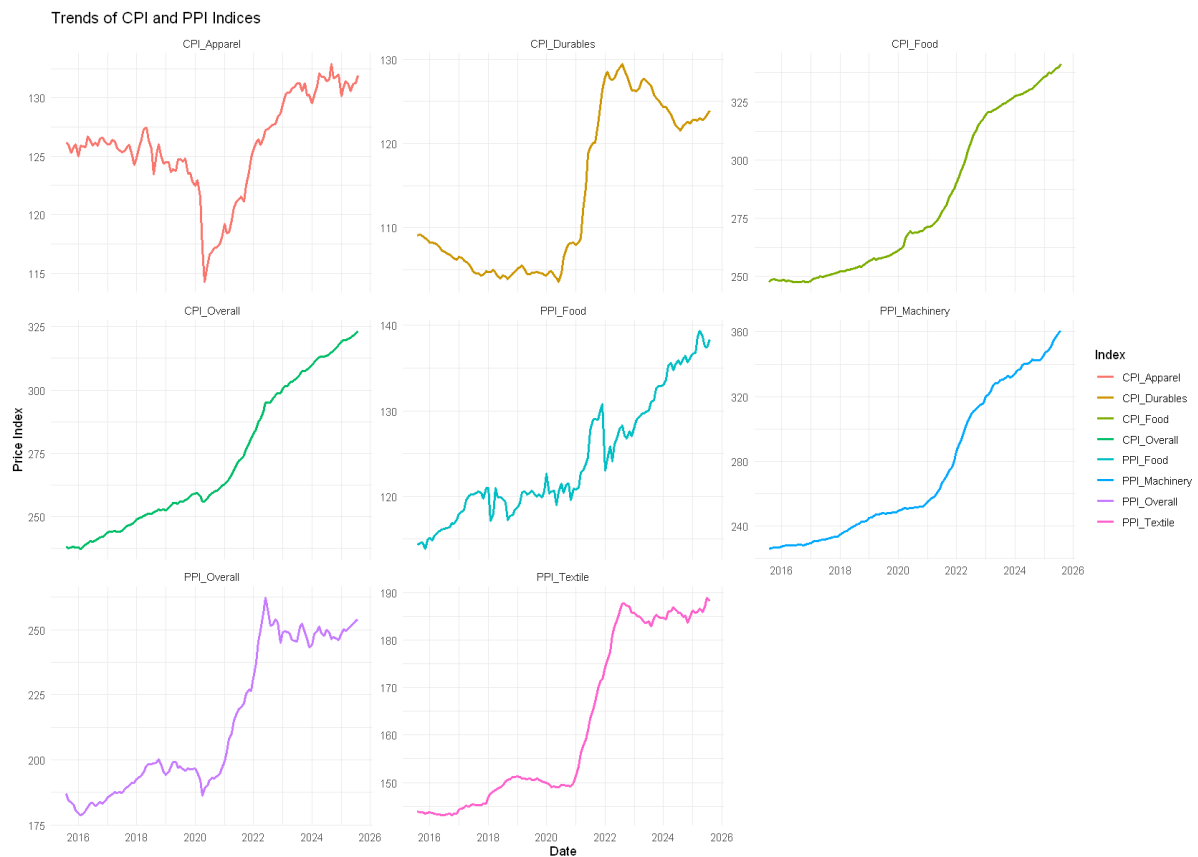


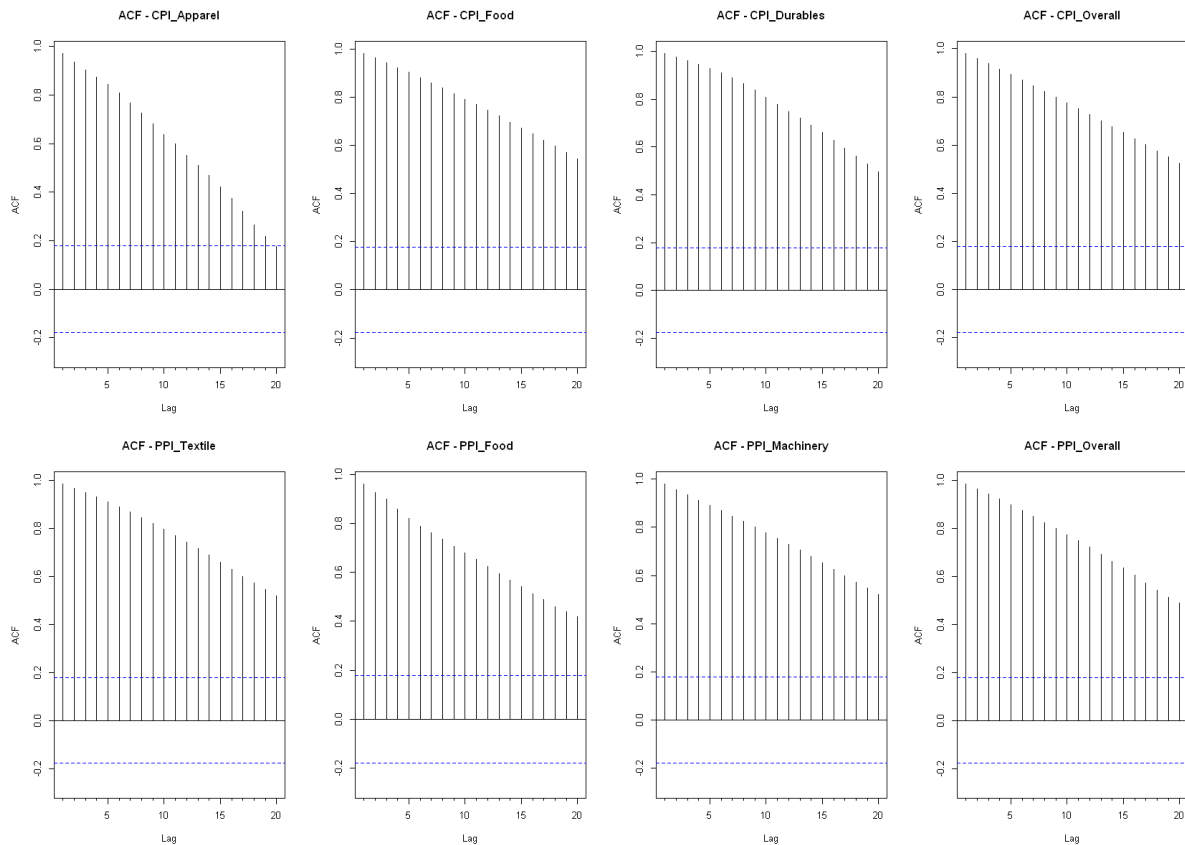
## Initial Time Series Exploratory Analysis

Before starting any analysis all the CPI and PPI data was plotted to see if there are any observable trends, seasonality or patterns in the data which could reduce the effectiveness of a time series model.



From the plot we can see that all the data series seem to have a general upward trend and there seems to be no seasonality exhibited in the plots.

The ACF plots of all the series were plotted to check for stationarity which is a key assumption for any time series models.

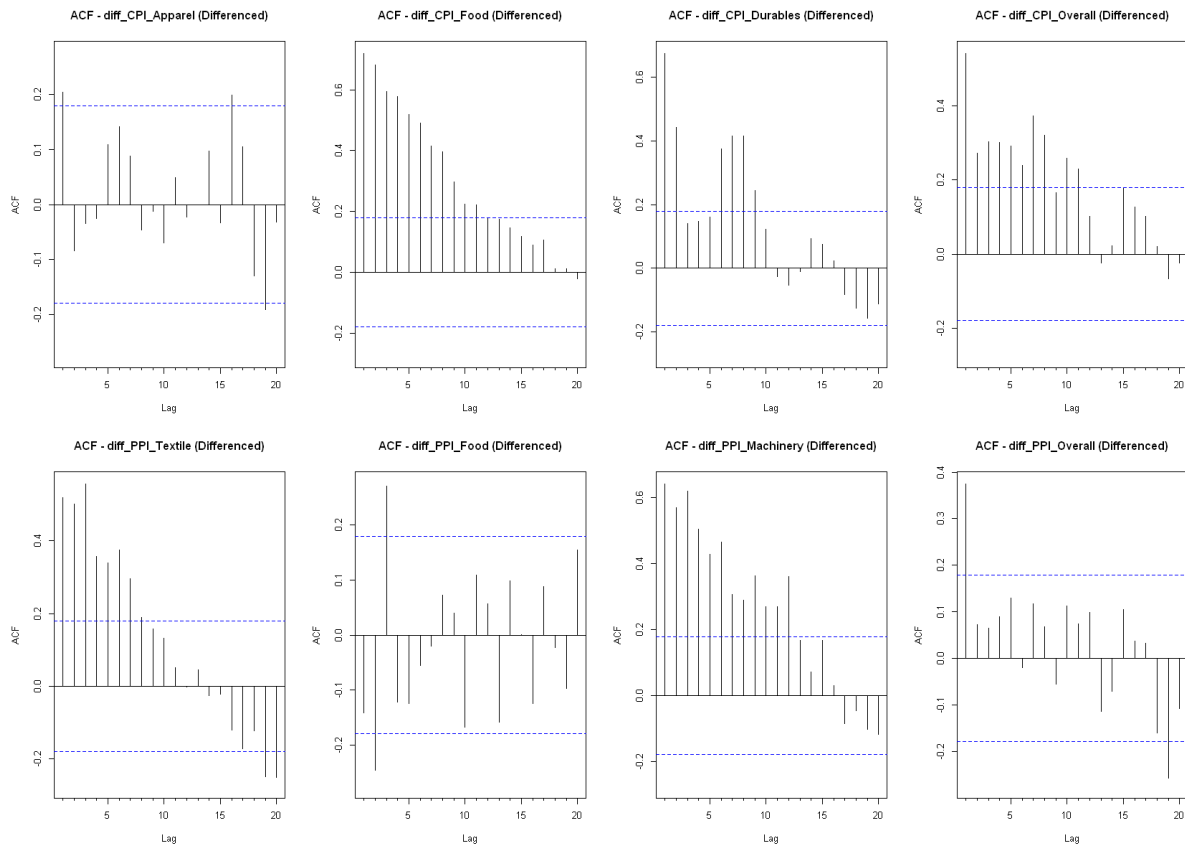


The ACF plots show a slow decay in autocorrelations for all the data series which is inline with the existence of trends that we observed visually. To formally test for stationarity the ADF (Augmented Dickey Fuller) test was applied to each series.

A data.frame: 8 × 2

|               | statistic.Dickey-Fuller | p_value   |
|---------------|-------------------------|-----------|
|               | <dbl>                   | <dbl>     |
| CPI_Apparel   | -1.316306               | 0.8605514 |
| CPI_Food      | -2.228079               | 0.4818294 |
| CPI_Durables  | -1.964756               | 0.5912056 |
| CPI_Overall   | -1.621521               | 0.7337746 |
| PPI_Textile   | -2.521774               | 0.3598377 |
| PPI_Food      | -2.060695               | 0.5513555 |
| PPI_Machinery | -2.000322               | 0.5764325 |
| PPI_Overall   | -1.914398               | 0.6121227 |

All of the tests gave a pvalue of  $P > 0.05$  so we fail to reject the null hypothesis meaning that all these series are non stationary. Such non stationarity needs to be addressed otherwise if we model the data directly using time series methods it may produce unreliable results. Hence, first order differencing was applied to each series then the ACF plots and ADF tests were applied to the differenced series.



|                                  | Series             | Statistic | P_Value    |
|----------------------------------|--------------------|-----------|------------|
| diff_CPI_Apparel.Dickey-Fuller   | diff_CPI_Apparel   | -4.307790 | 0.01000000 |
| diff_CPI_Food.Dickey-Fuller      | diff_CPI_Food      | -2.287327 | 0.45723800 |
| diff_CPI_Durables.Dickey-Fuller  | diff_CPI_Durables  | -2.982714 | 0.16842583 |
| diff_CPI_Overall.Dickey-Fuller   | diff_CPI_Overall   | -2.987580 | 0.16640504 |
| diff_PPI_Textile.Dickey-Fuller   | diff_PPI_Textile   | -2.656503 | 0.30390982 |
| diff_PPI_Food.Dickey-Fuller      | diff_PPI_Food      | -5.381645 | 0.01000000 |
| diff_PPI_Machinery.Dickey-Fuller | diff_PPI_Machinery | -2.528492 | 0.35707595 |
| diff_PPI_Overall.Dickey-Fuller   | diff_PPI_Overall   | -3.731801 | 0.02466632 |

From the ACF plots of Apparel CPI, Food PPI, Overall PPI we can see that apart from the first lag the autocorrelations are within the confidence band. This is confirmed with the results of the ADF test since the P values are  $<0.5$  for these series so we may be able to say these series are now weakly stationary after first order differencing. However, all the other series are still non stationary so they will need additional measures (either detrending using linear regression or log transformation before the differencing) to make the series stationary before modelling them.